# Federated Unlearning via Dual Distillation

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## Abstract

Federated Unlearning (FU) aims to remove the influence of target clients from a trained Federated Learning (FL) model, providing a practical alternative to costly retraining on the remaining clients. However, most existing FU methods require client re-participation or access to public data during the unlearning process, and they are often limited to single-client unlearning. These requirements significantly restrict the practical applicability of FU in real-world scenarios. To address these challenges, we propose Federated Unlearning via Dual Distillation (FU-Dual), a client-free FU framework that combines client-side dataset distillation during FL with server-side knowledge distillation during unlearning. Specifically, during the FL process, each client maintains a compact proxy dataset via dataset distillation, while the server collects these proxy datasets before the unlearning process. Based on the proxy datasets, the server learns two lightweight logit-space mappings, including an unlearning mapping to capture the effect of removing target clients, and a distillation mapping to model the discrepancy between real and proxy data. These two mappings bridge the “all-to-remaining” and “real-to-proxy” gaps, respectively, and are then frozen as dual teachers to guide knowledge distillation for obtaining the final unlearned model. Extensive experiments across diverse settings demonstrate that FU-Dual consistently outperforms existing baselines without requiring client re-participation, achieving strong post-unlearning accuracy while maintaining low Attack Success Rate (ASR) and Membership Inference Attack (MIA) accuracy close to random guessing.

## 1 Introduction

Federated learning (FL) enables collaborative model training across distributed clients without centralizing raw data, thereby preserving privacy in sensitive domains such as healthcare and finance [Kairouz et al., 2021, Sheller et al., 2020, Yang et al., 2019]. However, regulations such as the GDPR “right to be forgotten” introduce new requirements for FL systems, mandating the removal of user data and its influence from trained models [Regulation, 2018, Voigt and Von dem Bussche, 2017, Guo et al., 2020, Sekhari et al., 2021]. In practice, client-side data may contain unauthorized records or become legally restricted due to cross-border regulations or expired consent. Such data removal requests are common and consequential, particularly for clients operating under strict regulatory constraints. This brings increasing research on machine unlearning, which aims to remove the influence of specific data from trained models [Cao and Yang, 2015, Ginart et al., 2019].

The naive approach, retraining from scratch on cleansed data, guarantees exact unlearning but incurs prohibitive computational and communication costs, especially for FL with numerous training rounds [Liu et al., 2021]. This motivates federated unlearning (FU), which seeks to remove a client’s influence on the globally trained model while avoiding full retraining. However, federated unlearning (FU) introduces unique challenges beyond conventional machine unlearning. First, client data are locally stored and often privacy-sensitive, remaining decentralized and inaccessible to the server during the unlearning process. Second, unlearning requests may arise simultaneously from multiple clients, requiring the removal of their combined influence on the global model. Third, clients may not be able to participate in the unlearning process, introducing additional system-level constraints and complicating practical deployment.

Existing FU approaches can be categorized into three families [Romandini et al., 2024, Liu et al., 2024, Zhao et al., 2025]. *Replay-based methods* [Liu et al., 2021, Cao et al., 2023] store historical updates or checkpoints, but require substantial storage and raise privacy concerns. *Parameter editing methods* [Wu et al., 2022, Zhao et al., 2024, Khalil et al., 2025] directly modify model parameters through subtraction, momentum manipulation, or weight negation, but operate in the entangled parameter space, making client contributions difficult to isolate. *Distillation-based methods* use knowledge distillation techniques [Deng et al., 2024, Fu et al., 2024, Ye et al., 2024b], but require extra surrogate data to transfer knowledge.

The above FU methods reduce the overall cost compared to retraining from scratch, but rely heavily on clients' re-participation in the unlearning process. However, this assumption is often impractical in real-world deployments: clients may be offline, unavailable, or unwilling to engage, leading to significant delays and increased system complexity. Moreover, repeated client involvement further entangles model updates, making it difficult to isolate and remove individual client contributions in a principled manner. These limitations naturally raise the following key question: *How can we achieve effective federated unlearning without client re-participation?*

In this paper, we propose Federated Unlearning via Dual Distillation (FU-Dual), a novel server-side FU algorithm without client re-participation. To enable such a client-free design, we introduce a practical pre-instrumentation strategy, where each client that may request unlearning maintains a compact proxy dataset during the standard FL process. We assume that all clients may potentially request unlearning, ensuring general applicability. Our key insight is that unlearning can be decomposed into two objectives of removing target client influence and preserving model utility on remaining data. Instead of entangling these two objectives in parameter space, we learn two lightweight logit-space mappings, that is, an unlearning mapping that captures the effect of removing target clients, and a distillation mapping that bridges the real-proxy domain gap. Our main contributions are listed below.

- **Client-Free FU-Dual.** To the best of our knowledge, FU-Dual is the first client-free federated unlearning framework that supports multi-client unlearning without public dataset. We propose a dual distillation framework with dual logit-space mappings to achieve effective federated unlearning without client re-participation.
- **Dual Distillation.** FU-Dual integrates dataset distillation and knowledge distillation into a unified framework. During the FL phase, each client maintains a compact proxy dataset via dataset distillation. During the unlearning phase, the server performs knowledge distillation on these proxy datasets to obtain the unlearned model, without requiring client re-participation or auxiliary public data.
- **Dual Mappings.** We design two lightweight logit-space mappings as complementary teachers for knowledge distillation. The unlearning mapping captures the effect of removing target clients, thereby bridging the "all-to-remaining" gap, while the distillation mapping models the discrepancy between real and proxy data, thereby bridging the "real-to-proxy" gap. Together, these mappings provide stable and effective supervision for client-free federated unlearning.
- **Extensive Experiments.** We comprehensively evaluate FU-Dual across various datasets, model architectures, non-IID degrees, and numbers of unlearned clients. Experimental results show that FU-Dual consistently outperforms existing federated unlearning methods, achieving high post-unlearning accuracy while keeping the Attack Success Rate (ASR) low and Membership Inference Attack (MIA) accuracy close to random guessing.

## 2 Related Work

Federated unlearning (FU) has been studied as an efficient alternative to retraining on remaining clients, with recent surveys grouping existing methods into replay-based, parameter-editing, and distillation-based approaches [Romandini et al., 2024, Liu et al., 2024, Zhao et al., 2025].

Replay-based methods, such as FedEraser, FedRecover, FedADP, and Crab [Liu et al., 2021, Cao et al., 2023, Jiang et al., 2024, 2025], approximate the training trajectory after removing target clients by storing checkpoints or historical updates. They reduce retraining cost, but their accuracy depends on how faithfully the cached trajectory represents the original FL dynamics, and recovery often requires additional interaction with remaining clients. Parameter-editing methods instead modify model weights or update statistics directly, including contribution subtraction, momentum degradation, and weight negation [Zhang et al., 2023, Guo et al., 2023, Zhao et al., 2024, Khalil et al., 2025]. These methods are lightweight, but client effects are entangled in parameter space, so direct edits can damage useful representations and may need post-unlearning repair.

Table 1: **FU Methods Contrast.**

| **Method** | **Unlearn client-free** | **Remain client-free** | **Public data-free** | **Multi-client unlearn** |
|---|---|---|---|---|
| FCU | ✗ | ✓ | ✓ | ✓ |
| FedEraser | ✓ | ✗ | ✓ | ✓ |
| FUKD | ✓ | ✓ | ✗ | ✓ |
| CFU | ✓ | ✓ | ✗ | ✓ |
| FFMU | ✓ | ✓ | ✓ | ✗ |
| PGD | ✗ | ✓ | ✓ | ✗ |
| MoDe | ✓ | ✗ | ✓ | ✗ |
| **FU-Dual** | ✓ | ✓ | ✓ | ✓ |

Another line uses distillation, training reconstruction, or synthetic supervision to preserve utility after forgetting [Wu et al., 2022, Deng et al., 2024, Fu et al., 2024, Ye et al., 2024a,b]. Client-Free FU (CFU) [Fu et al., 2024] is particularly related because it also targets client-free unlearning, using training reconstruction with anchor subspace calibration to avoid client participation at deletion time, although it depends on auxiliary data for calibration and recovery. FFMU [Che et al., 2023] instead moves unlearning preparation into federated training through smooth-quantized gradient updates motivated by nonlinear functional theory, so the deletion stage requires no additional client-side optimization. Related dataset-distillation methods construct compact proxies for real data [Zhao and Bilen, 2023, Xiong et al., 2023, Zhao et al., 2023, Sun et al., 2024], suggesting a practical way to avoid storing raw client data.

Most existing FU methods require the re-participation of unlearned or remaining clients, or depend on auxiliary public datasets, or are limited to single-client unlearning. These requirements restrict their applicability in practical scenarios where clients may be unavailable, public data may be inaccessible, and multiple clients may request unlearning simultaneously. In contrast, FU-Dual performs client-free unlearning without relying on public data, while supporting multi-client unlearning in a unified framework. Table 1 summarizes the comparison between FU-Dual and existing baselines.

## 3 Motivation

While the server has no access to the clients' real data, we propose to leverage proxy datasets to enable server-side federated unlearning without client re-participation. This naturally raises two key questions in developing FU-Dual, with Key notations summarized in Table 2 to facilitate the following discussion.

- How can we effectively utilize the proxy data?
- How can we effectively obtain the proxy data?

Table 2: **Summary of notations.**

| **Symbol** | **Description** |
|---|---|
| $\mathcal{T}_U(\cdot;\phi)$ | Unlearning map: $f(X;\hat{\theta}_{\mathcal{N}}) \to f(X;\hat{\theta}_{\mathcal{R}})$ |
| $\mathcal{T}_D(\cdot;\psi)$ | Distillation map: $f(X;\hat{\theta}_{\mathcal{N}}) \to f(X;\theta_{\mathcal{N}})$ |
| $f(X;\theta)$ | Output of model $\theta$ on dataset $X$ |
| $\mathcal{D}_{\mathcal{N}}$, $\theta_{\mathcal{N}}$ | All-client real data / global FL model |
| $\mathcal{D}_{\mathcal{R}}$, $\theta_{\mathcal{R}}$ | Remaining-client real data / target model |
| $\hat{\mathcal{D}}_{\mathcal{N}}$, $\hat{\theta}_{\mathcal{N}}$ | All-client proxy data / ref. model |
| $\hat{\mathcal{D}}_{\mathcal{R}}$, $\hat{\theta}_{\mathcal{R}}$ | Remaining-client proxy data / ref. model |

For the first question, we initially consider an ideal setting without privacy constraints, where the proxy dataset $\hat{\mathcal{D}}_i$ is identical to the real dataset $\mathcal{D}_i$ (i.e., $\hat{\mathcal{D}}_i = \mathcal{D}_i$). Under this ideal setting, the unlearning problem reduces to training from scratch on the remaining real data $\mathcal{D}_{\mathcal{R}} = \{\mathcal{D}_i \mid i \in \mathcal{R}\}$, which yields the optimal unlearned model $\theta_{\mathcal{R}}$.

Notably, for any dataset $\mathcal{D} = (X, Y)$, the model outputs of $f(X;\theta_{\mathcal{R}})$ and $f(X;\theta_{\mathcal{N}})$ are closely related. This relationship is particularly transparent when $f(\cdot)$ is linear, and the effect of removing target clients can be characterized as a structured transformation in the output (logit) space.

Based on this observation, we define an ideal unlearning mapping $\mathcal{T}_U$ such that

$$\mathcal{T}_U^* : f(X;\theta_{\mathcal{N}}) \to f(X;\theta_{\mathcal{R}}),$$

which captures the output-level transformation induced by removing the target clients. In other words, the unlearned model can be obtained if such an unlearning mapping can be identified.

**Lemma 3.1** (Existence of the Unlearning Mapping)**.** *Let $f(X;\theta) = X\theta$ be a linear model, and let $\hat{\mathcal{D}}_i = \mathcal{D}_i$, we have the ideal unlearning mapping as*

$$\begin{aligned} f(X;\theta_{\mathcal{R}}) &= \mathcal{T}_U^*(f(X;\theta_{\mathcal{N}});\phi^*) = \phi^* f(X;\theta_{\mathcal{N}}) \\ \phi^* &= \theta_{\mathcal{N}}^{\dagger}\theta_{\mathcal{R}} \end{aligned} \tag{1}$$

*where $\theta_{\mathcal{N}}^{\dagger}$ denotes the Moore–Penrose pseudoinverse of $\theta_{\mathcal{N}}$.*

Furthermore, we assume that the proxy dataset is sufficiently representative of the real data, for example, as a subset such that $\hat{\mathcal{D}}_i \subset \mathcal{D}_i$. This implies $\hat{\mathcal{D}}_{\mathcal{R}} \subset \mathcal{D}_{\mathcal{R}}$ and $\hat{\mathcal{D}}_{\mathcal{N}} \subset \mathcal{D}_{\mathcal{N}}$. Based on these proxy datasets, we can train the corresponding surrogate models $\hat{\theta}_{\mathcal{N}}$ and $\hat{\theta}_{\mathcal{R}}$. This naturally induces another unlearn mapping $\mathcal{T}_U$, such that

$$\mathcal{T}_U(\cdot;\phi) : f(X;\hat{\theta}_{\mathcal{N}}) \to f(X;\hat{\theta}_{\mathcal{R}}).$$

We then want to capture the distance between the surrogate mapping $\phi$ and the ideal mapping $\phi^*$.

**Theorem 3.2** (Bounded Error of Unlearning Mapping)**.** *Let $\hat{\mathcal{D}}_i \subset \mathcal{D}_i$, and let $\theta_i, \hat{\theta}_i$ be the surrogate models trained on $\mathcal{D}_i, \hat{\mathcal{D}}_i$, respectively. We have*

$$\mathbb{E}\big\|\phi - \phi^*\big\|_2^2 \le \xi \left[\sum_{i\in\mathcal{N}} \mathbb{E}\|\theta_i - \hat{\theta}_i\|_2^2 + \sum_{i\in\mathcal{R}} \mathbb{E}\|\theta_i - \hat{\theta}_i\|_2^2\right] \tag{2}$$

$$\le \xi \left[\sum_{i\in\mathcal{N}} \mathbb{E}\|\frac{f(X;\theta_i) - f(X;\hat{\theta}_i)}{\|X\|_2^2}\|_2^2 + \sum_{i\in\mathcal{R}} \mathbb{E}\|\frac{f(X;\theta_i) - f(X;\hat{\theta}_i)}{\|X\|_2^2}\|_2^2\right] \tag{3}$$

$$\le O(\frac{1-\rho}{\rho}) \tag{4}$$

*where $\rho$ is bounded by the sampling ratio, and $\xi$ denotes a positive constant.*

Although the condition $\hat{\mathcal{D}}_i \subset \mathcal{D}_i$ is generally unattainable in practice, we assume for analytical tractability that there exists a proxy dataset $\hat{\mathcal{D}}_i$ satisfy the following assumption.

**Assumption 3.3.** There exists a proxy dataset $\hat{\mathcal{D}}_i$ and a positive constant $C_i > 0$ satisfying, for all $x$,

$$\left\|\frac{f(X;\theta_i) - f(X;\hat{\theta}_i)}{\|X\|_2^2}\right\|_2 \le C_i.$$

**Proposition 3.4** (Bounded Error of Unlearning Mapping)**.** *Let the proxy data $\hat{\mathcal{D}}_i$ satisfy Assumption 3.3, and let $\theta_i, \hat{\theta}_i$ be the surrogate models trained on $\mathcal{D}_i, \hat{\mathcal{D}}_i$, respectively. We have*

$$\mathbb{E}\big\|\phi - \phi^*\big\|_2^2 \le \xi \left[\sum_{i\in\mathcal{N}} C_i^2 + \sum_{i\in\mathcal{R}} C_i^2\right]. \tag{5}$$

Similarly, there exist a distillation mapping $\mathcal{T}_D : f(X;\hat{\theta}_{\mathcal{N}}) \to f(X;\theta_{\mathcal{N}})$ with bounded error to bridge the gap between proxy data and real data in terms of their induced model outputs. All proof details refer to Appendix A.

For the second question, we note that the goal of obtaining proxy data is not to recover the real data itself, but to ensure that the resulting model on the proxy data exhibits behavior similar to a model trained on the real data. This observation motivates us to leverage dataset distillation to construct the proxy data. Dataset distillation aims to generate a compact synthetic dataset that serves as a surrogate for the real dataset, such that a model trained on the distilled data exhibits behavior similar to one trained on the full dataset. The general objective of dataset distillation can be formulated as in Yu et al. [2024]

$$\min_{\hat{\mathcal{D}}_i} \int p_i(x)\, d\Big(f(x;\theta_i), f(x;\hat{\theta}_i)\Big)\ dx, \tag{6}$$

where $p_i(x)$ denotes the class-conditional local distribution of client $i$; for samples with label $y$, we write $p_i(x) \triangleq p_i(x \mid y)$. The above optimization objective encourages the generated distilled dataset $\hat{\mathcal{D}}_i$ to satisfy Assumption 3.3.

**Privacy of proxy dataset.** The distilled proxy dataset are privacy-preserving in our empirical evaluation: their membership leakage remains close to random guessing (AUC 0.498–0.510, TPR@1%FPR 0.010–0.013), while their FID scores (93.79–236.66) indicate a clear distributional separation from the original data. Full privacy measurements and qualitative examples are reported in Appendix D.

# 4 Methodology

**4.1 Overview** FU-Dual innovatively realizes client-free federated unlearning by combining data distillation and knowledge distillation, without client re-participation or public data. As a compromise, each client maintains a proxy dataset during the federated learning phase for potential unlearning requests. After collecting these proxy datasets from clients, the unlearning process is performed entirely on the server side. The overall framework of FU-Dual is illustrated in Figure 1, with its key components described below.

Algorithm 1: FU-Dual

**Inputs:** $\theta_{\mathcal{N}}$, $\{\hat{\mathcal{D}}_i\}_{i\in\mathcal{N}}$, training round $T$
**Output:** target unlearn parameters $\theta_{\mathcal{R}}$
**Initialize:** target unlearn parameters $\theta_{\mathcal{R}}^0$

$\hat{\mathcal{D}}_{\mathcal{N}} \leftarrow \cup_{i\in\mathcal{N}} \hat{\mathcal{D}}_i$ ▷ Aggregate all proxy data
$\hat{\mathcal{D}}_{\mathcal{R}} \leftarrow \cup_{i\in\mathcal{R}} \hat{\mathcal{D}}_i$ ▷ Aggregate remain proxy data
$\hat{\theta}_{\mathcal{N}} \leftarrow$ Train model on $\hat{\mathcal{D}}_{\mathcal{N}}$
$\hat{\theta}_{\mathcal{R}} \leftarrow$ Train model on $\hat{\mathcal{D}}_{\mathcal{R}}$
$f_{\mathcal{N}} \leftarrow f(\hat{\mathcal{D}}_{\mathcal{R}}; \theta_{\mathcal{N}})$ ▷ logits of $\theta_{\mathcal{N}}$ on $\hat{\mathcal{D}}_{\mathcal{R}}$
$f_{\hat{\mathcal{N}}} \leftarrow f(\hat{\mathcal{D}}_{\mathcal{R}}; \hat{\theta}_{\mathcal{N}})$ ▷ logits of $\hat{\theta}_{\mathcal{N}}$ on $\hat{\mathcal{D}}_{\mathcal{R}}$
$f_{\hat{\mathcal{R}}} \leftarrow f(\hat{\mathcal{D}}_{\mathcal{R}}; \hat{\theta}_{\mathcal{R}})$ ▷ logits of $\hat{\theta}_{\mathcal{R}}$ on $\hat{\mathcal{D}}_{\mathcal{R}}$
$\mathcal{T}_{\mathcal{U}} \leftarrow \mathrm{MLP}(f_{\hat{\mathcal{N}}} \rightarrow f_{\hat{\mathcal{R}}})$ ▷ Unlearn mapping
$\mathcal{T}_{\mathcal{D}} \leftarrow \mathrm{MLP}(f_{\hat{\mathcal{N}}} \rightarrow f_{\mathcal{N}})$ ▷ Distillation mapping
**for** $t = 1, \ldots, T$ **do**
  Sample mini-batch $B$ from $\hat{\mathcal{D}}_{\mathcal{R}}$
  $f_{\hat{\mathcal{N}}}^{\mathcal{T}_{\mathcal{U}}}(B) \leftarrow \mathcal{T}_{\mathcal{U}}(f_{\hat{\mathcal{N}}}(B), \phi)$ ▷ Unlearn teacher
  $f_{\hat{\mathcal{R}}}^{\mathcal{T}_{\mathcal{D}}}(B) \leftarrow \mathcal{T}_{\mathcal{D}}(f_{\hat{\mathcal{R}}}(B), \psi)$ ▷ Distillation teacher
  $\mathcal{L}(B) \leftarrow \mathcal{L}(f_{\mathcal{R}}^s(B), f_{\hat{\mathcal{N}}}^{t_U}(B), f_{\hat{\mathcal{R}}}(B))$ ▷ By (13)
  $\theta_{\mathcal{R}}^{t+1} \leftarrow \theta_{\mathcal{R}}^t - \eta \nabla_{\theta_{\mathcal{R}}} \mathcal{L}(B)$
**end for**

**Data Distillation for Proxy Dataset.** In the FL phase, each client who may participate in the unlearning process is required to maintain a proxy dataset. Once the unlearning process begins, each client only needs to upload this proxy dataset to the server, without needing to re-participate in the subsequent unlearning process.

**Unlearning and Distillation Mappings.** After collecting the proxy datasets from clients, the server constructs both an unlearning mapping and a distillation mapping in the logit space. Specifically, the unlearning mapping captures the logits transition before and after the unlearning stage, while the distillation mapping captures the logits transition between the real data and the distilled proxy dataset.

**Knowledge Distillation for Model Unlearning.** The aforementioned unlearning and distillation Mappings serve as the teacher models in knowledge distillation to guide the unlearning process on the server-side.

**4.2 Data Distillation for Proxy Dataset** In addition to the standard FL process, each client participating in FU maintains a distilled proxy dataset for subsequent unlearning. Section 3 justifies why such a proxy can replace raw local data: the dataset-distillation objective in Eq. (6) encourages the model trained on $\hat{\mathcal{D}}_i$ to match the output behavior of the model trained on $\mathcal{D}_i$. Therefore, this subsection focuses on the preparation stage of FU-Dual, i.e., how the proxy data are produced and what intermediate artifacts are made available before the mapping stage.

Specifically, during federated training, client $i$ trains on its private dataset $\mathcal{D}_i$, while the server aggregates client updates to obtain the all-client FL model $\theta_{\mathcal{N}}$. In parallel, client $i$ optimizes a compact distilled dataset $\hat{\mathcal{D}}_i$ according to Eq. (6), where $\theta_i$ and $\hat{\theta}_i$ denote the same model architecture trained on $\mathcal{D}_i$ and $\hat{\mathcal{D}}_i$, respectively. The synthetic samples inherit the labels of the local classes, and their size is controlled by the distillation budget, e.g., images per class (IPC), rather than by the size of the original local dataset. Thus, the client-side product of this stage is only a compact labeled proxy set $\hat{\mathcal{D}}_i = (\hat{X}_i, \hat{Y}_i)$, which is much smaller than $\mathcal{D}_i$ and avoids retaining raw local samples for later unlearning.

Once these proxy sets are available to the server, no further client-side computation is needed. The server only organizes them into two collections:

$$\hat{\mathcal{D}}_{\mathcal{N}} = \bigcup_{i\in\mathcal{N}} \hat{\mathcal{D}}_i, \qquad \hat{\mathcal{D}}_{\mathcal{R}} = \bigcup_{i\in\mathcal{R}} \hat{\mathcal{D}}_i. \tag{7}$$

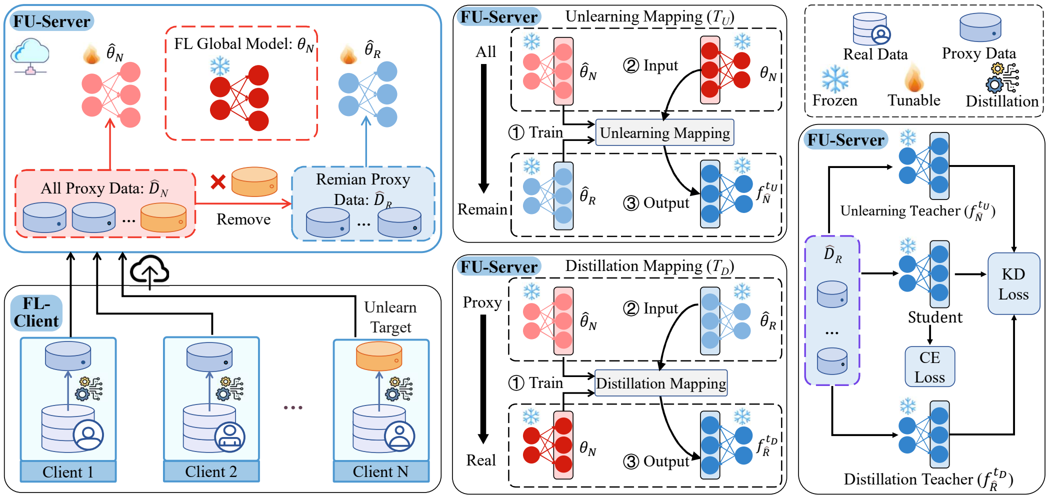



Figure 1: **Overview of the FU-Dual framework.** In the lower-left Client Side panel, clients generate proxy datasets during FL and upload them upward to the Server Side. From left to right on the Server Side, the FU-Server obtains the global model, trains reference models, constructs unlearning/distillation mappings, and distills knowledge to obtain the unlearned model.

It then trains the same architecture on $\hat{\mathcal{D}}_\mathcal{N}$ and $\hat{\mathcal{D}}_\mathcal{R}$ to obtain the proxy-domain reference models $\hat{\theta}_\mathcal{N}$ and $\hat{\theta}_\mathcal{R}$. Thus, this stage provides the FL model $\theta_\mathcal{N}$, the distilled data $\{\hat{\mathcal{D}}_i\}$, and the proxy reference models; the logit mappings are learned later in Section 4.3.

**4.3 Unlearning and Distillation Mappings** We formulate the unlearning process as knowledge distillation from the global parameters $\theta_\mathcal{N}$ (trained on all clients) to unlearned parameters $\theta_\mathcal{R}$ that behave as trained on the remaining clients. We then construct two mappings to obtain the effective teacher signal for the KD process on the proxy remaining dataset $\hat{\mathcal{D}}_\mathcal{R} = (\hat{X}_R, \hat{Y}_R)$, after collecting proxy datasets $\hat{\mathcal{D}}_i$ from all clients.

**Unlearning mapping $\mathcal{T}_\mathcal{U}$.** Because raw $\mathcal{D}_\mathcal{R}$ is unavailable, Eq. (8) learns a proxy map from $\hat{\theta}_\mathcal{N}$ logits to $\hat{\theta}_\mathcal{R}$ logits.

$$\min_{\phi} \mathbb{E}\|\mathcal{T}_\mathcal{U}(f(\hat{\mathcal{D}}_\mathcal{R}; \hat{\theta}_\mathcal{N}), \phi) - f(\hat{\mathcal{D}}_\mathcal{R}; \hat{\theta}_\mathcal{R})\|_1. \tag{8}$$

This unlearning mapping captures the unlearning effect by transforming logits from the "all-clients" state to the "remaining-clients" state, thereby bridging the "all-remain" gap in the proxy domain.

**Distillation mapping $\mathcal{T}_\mathcal{D}$.** Besides the above "all-remain" gap, the proxy and real domains also differ. Since the server has access to the original FL parameters $\theta_\mathcal{N}$, we learn $\mathcal{T}_D : f(\hat{\mathcal{D}}_\mathcal{R}; \hat{\theta}_\mathcal{N}) \to f(\hat{\mathcal{D}}_\mathcal{R}; \theta_\mathcal{N})$ to bridge this "real-proxy" gap by minimizing Eq. (9).

$$\min_{\psi} \mathbb{E}\|\mathcal{T}_\mathcal{D}(f(\hat{\mathcal{D}}_\mathcal{R}; \hat{\theta}_\mathcal{N}), \psi) - f(\hat{\mathcal{D}}_\mathcal{R}; \theta_\mathcal{N})\|_1. \tag{9}$$

This mapping transfers proxy-domain logits to the real-domain behavior, providing a utility-preserving teacher signal for subsequent unlearning.

After being trained on the proxy remaining dataset $\hat{\mathcal{D}}_\mathcal{R}$ via Eqs. (8) and (9), both unlearning mapping $\mathcal{T}_\mathcal{U}(\cdot, \phi)$ and distillation mapping $\mathcal{T}_\mathcal{D}(\cdot, \psi)$ are frozen to collectively serve as the teacher signal for the subsequent knowledge distillation to realize federated unlearning.

**4.4 Knowledge Distillation for Model Unlearning** With the two mappings learned in Eqs. (8) and (9) as the teacher signal, we are eventually able to obtain the unlearned model $\theta_\mathcal{R}$ by knowledge distillation on the proxy remaining dataset $\hat{\mathcal{D}}_\mathcal{R}$.

The desired unlearn model $\theta_\mathcal{R}$ is taken as the student model, and its logit output is defined in Eq. (10).

$$f_\mathcal{R}^s = f(\hat{\mathcal{D}}_\mathcal{R}; \theta_\mathcal{R}). \tag{10}$$

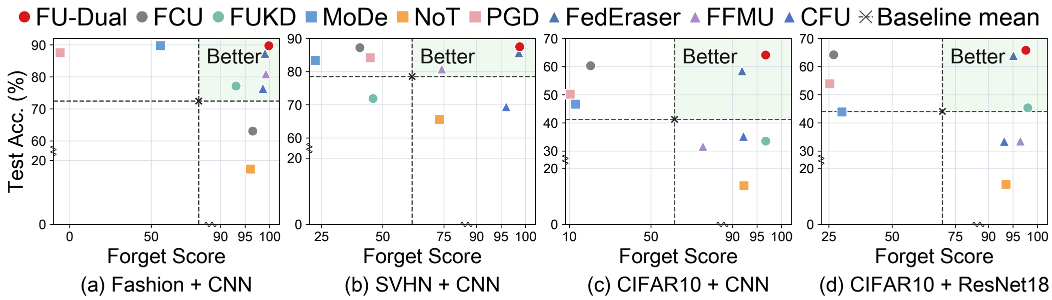



Figure 2: **Accuracy and Forget Score.** High accuracy and low Forget Score are better, where Forget Score = ASR + |MIA - 50|.

Based on the unlearning mapping $\mathcal{T}_{\mathcal{U}}(\cdot, \phi)$, we construct the unlearning teacher signal in Eq. (11) to bridge the "all-remain" gap.

$$f_{\hat{\mathcal{N}}}^{\mathcal{T}_{\mathcal{U}}} = \mathcal{T}_{\mathcal{U}}(f(\hat{\mathcal{D}}_{\mathcal{R}}; \theta_{\mathcal{N}}), \phi). \tag{11}$$

Similarly, based on the distillation mapping $\mathcal{T}_{\mathcal{D}}(\cdot, \psi)$, we construct the distillation teacher signal in Eq. (12) to bridge the "real-proxy" gap.

$$f_{\hat{\mathcal{R}}}^{\mathcal{T}_{\mathcal{D}}} = \mathcal{T}_{\mathcal{D}}(f(\hat{\mathcal{D}}_{\mathcal{R}}; \hat{\theta}_{\mathcal{R}}), \psi). \tag{12}$$

The two teacher signals in Eqs. (11) and (12) guide the training of the unlearn model $\theta_{\mathcal{R}}$, together with the conventional cross-entropy loss. The final objective is given in Eq. (13).

$$\mathcal{L} = \mathcal{L}_{CE}(\hat{Y}_R, f_{\mathcal{R}}^s) + \alpha\, \mathcal{L}_{KD}(f_{\hat{\mathcal{N}}}^{t_U}, f_{\mathcal{R}}^s) + \beta\, \mathcal{L}_{KD}(f_{\hat{\mathcal{R}}}^{\mathcal{T}_{\mathcal{D}}}, f_{\mathcal{R}}^s), \tag{13}$$

where $\hat{Y}_R$ is the ground-truth label of $\hat{\mathcal{D}}_{\mathcal{R}}$, $\alpha$ and $\beta$ are nonnegative parameters to balance the above three supervision signals.

Notably, the dual-teacher signals in Eqs. (11) and (12) provide effective guidance for training the unlearn model $\theta_R$, where $f_{\hat{\mathcal{N}}}^{\mathcal{T}_{\mathcal{U}}}$ and $f_{\hat{\mathcal{R}}}^{\mathcal{T}_{\mathcal{D}}}$ bridge the "all-remain" and "real-proxy" gaps, respectively.

# 5 Experiments

**5.1 Experimental Setup** We evaluate **FU-Dual** on Fashion-MNIST, SVHN, and CIFAR-10 under non-IID federated client unlearning. We use CNN models for all three datasets and ResNet-18 for CIFAR-10. We report test accuracy for utility, attack success rate (ASR) and membership inference attack (MIA) accuracy for forgetting, and communication/computation costs.

We compare with Retrain, FedEraser [Liu et al., 2021], FUKD [Wu et al., 2022], FCU [Deng et al., 2024], CFU [Fu et al., 2024], FFMU [Che et al., 2023], PGD [Halimi et al., 2022], MoDe [Zhao et al., 2024], and NoT [Khalil et al., 2025]. All setting details, including data partitioning, training, distillation, metrics, and hardware, are provided in Appendix H.

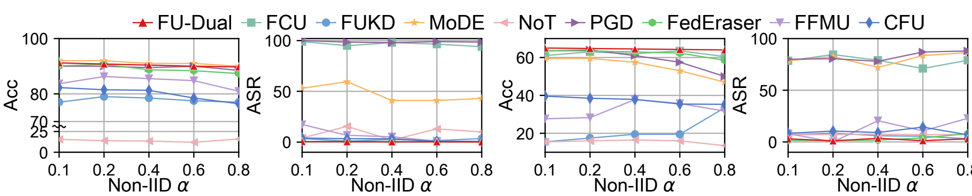
## 5.2 Comparison Study

**Strong overall performance.** Table 3 compares methods across datasets, forgetting metrics, and unlearning-stage costs, while Fig.2 provides a visual summary. Overall, **FU-Dual** achieves strong performance, with competitive or best unlearning accuracy, low ASR, and MIA values close to 50%, indicating effective forgetting with preserved utility. Although MoDe attains higher test accuracy on FashionMNIST, it also yields a substantially higher ASR, suggesting less complete forgetting. More detailed baseline results and variances across multiple runs are provided in AppendixE.

**Unlearning-stage cost.** FU-Dual performs post-request unlearning without client re-participation, so its client-side communication and computation costs are both zero in Table 3. Its server-side cost is also competitive with other methods, except for NOT, whose objective is to obtain a better model initialization rather than performing standard post-request unlearning. More detailed computation accounting is provided in Appendix G.

Table 3: **Main results across datasets and architectures.** Best results are **bold**, and second best are underlined (excluding Retrain). The full table including CFU and FFMU is provided in Appendix E.

| Dataset + Model | Method | Test Acc ↑ | ASR ↓ | MIA | Server Cost Comm. | Server Cost Comp. | Client Cost Comm. | Client Cost Comp. |
|---|---|---|---|---|---|---|---|---|
| FashionMNIST + CNN | Retrain | 92.16% | 0% | 50.03% | 1.72e8 | 0 | 1.72e8 | 2.35e15 |
| | FedEraser | 87.36% | 0.43% | 50.52% | 6.86e7 | 0 | 6.86e7 | 1.31e15 |
| | FUKD | 77.15% | 3.21% | 53.66% | 1.72e8 | 1.74e13 | 1.72e8 | 3.54e15 |
| | FCU | 63.09% | 3.14% | 50.31% | 1.86e8 | 0 | 1.84e8 | 4.35e14 |
| | PGD | 87.58% | 98.47% | 57.43% | 2.54e6 | 0 | 1.27e6 | 1.30e13 |
| | MoDe | **89.78%** | 41.20% | 53.12% | 3.60e8 | 0 | 3.49e8 | 5.00e15 |
| | NoT | 17.32% | 3.16% | 49.26% | 1.14e7 | 0 | 1.14e7 | 0 |
| | **FU-Dual (Ours)** | 89.74% | **0.15%** | **50.02%** | 0 | 4.70e13 | 0 | 0 |
| SVHN + CNN | Retrain | 89.97% | 0% | 50.02% | 1.73e8 | 0 | 1.73e8 | 3.01e15 |
| | FedEraser | 85.73% | **1.06%** | 50.31% | 6.91e7 | 0 | 6.91e7 | 1.67e15 |
| | FUKD | 71.91% | 50.63% | 53.36% | 1.73e8 | 4.76e13 | 1.73e8 | 4.55e15 |
| | FCU | 87.23% | 56.22% | 53.15% | 1.87e8 | 0 | 1.86e8 | 5.57e14 |
| | PGD | 84.23% | 51.98% | 53.20% | 2.56e6 | 0 | 1.28e6 | 1.60e13 |
| | MoDe | 83.41% | 74.35% | 53.24% | 3.62e8 | 0 | 3.52e8 | 6.41e15 |
| | NoT | 65.65% | 23.72% | 53.08% | 1.15e7 | 0 | 1.15e7 | 0 |
| | **FU-Dual (Ours)** | **87.52%** | 1.07% | **50.13%** | 0 | 6.10e13 | 0 | 0 |
| CIFAR10 + CNN | Retrain | 72.21% | 0% | 50.00% | 1.73e8 | 0 | 1.73e8 | 2.06e15 |
| | FedEraser | 58.52% | 7.98% | 50.09% | 6.91e7 | 0 | 6.91e7 | 1.14e15 |
| | FUKD | 33.55% | **3.17%** | **50.06%** | 1.73e8 | 1.83e13 | 1.73e8 | 3.09e15 |
| | FCU | 60.32% | 78.81% | 50.81% | 1.87e8 | 0 | 1.86e8 | 3.80e14 |
| | PGD | 50.23% | 88.54% | 51.09% | 2.56e6 | 0 | 1.28e6 | 1.10e13 |
| | MoDe | 46.68% | 87.00% | 50.09% | 3.62e8 | 0 | 3.52e8 | 4.37e15 |
| | NoT | 13.76% | 7.39% | 49.71% | 1.15e7 | 0 | 1.15e7 | 0 |
| | **FU-Dual (Ours)** | **64.13%** | **3.17%** | 50.08% | 0 | 6.00e13 | 0 | 0 |
| CIFAR10 + ResNet18 | Retrain | 76.33% | 0% | 50.01% | 2.68e10 | 0 | 2.68e10 | 1.14e16 |
| | FedEraser | 63.97% | 4.23% | 50.71% | 1.34e10 | 0 | 1.34e10 | 5.70e15 |
| | FUKD | 45.37% | **1.90%** | 50.10% | 0 | 6.66e13 | 6.70e8 | 0 |
| | FCU | 64.21% | 68.01% | 55.12% | 4.47e7 | 0 | 4.47e7 | 1.11e15 |
| | PGD | 53.88% | 71.52% | 53.14% | 8.94e7 | 0 | 4.47e7 | 5.55e13 |
| | MoDe | 43.91% | 69.22% | 50.63% | 2.23e9 | 0 | 2.23e9 | 5.88e14 |
| | NoT | 14.31% | 5.35% | 48.91% | 4.47e7 | 0 | 0 | 0 |
| | **FU-Dual (Ours)** | **65.82%** | 2.37% | **50.02%** | 0 | 5.44e13 | 0 | 0 |

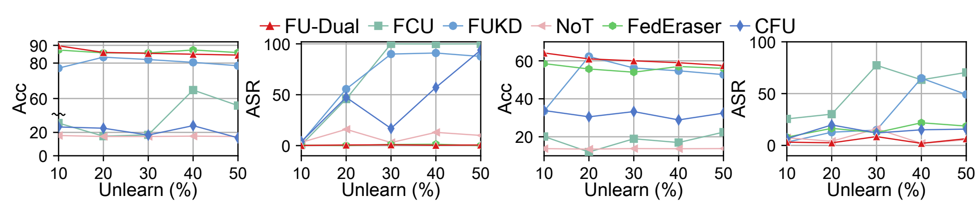


(a) Fashion + Acc (b) Fashion + ASR (c) CIFAR-10 + Acc (d) CIFAR-10 + ASR

Figure 3: Accuracy and ASR under different non-IID degrees on Fashion-MNIST and CIFAR-10.

**5.3 Robustness Analysis** We next test whether the accuracy–forgetting advantage persists when the data distribution, deletion demand, federation size, and proxy-distillation budget change.

**Robustness to data heterogeneity.** Fig. 3 evaluates robustness under varying non-IID degrees on Fashion-MNIST and CIFAR-10. **FU-Dual** keeps a stable accuracy–ASR balance as heterogeneity changes, while competing methods often trade one metric for the other.



(a) Fashion + Acc (b) Fashion + ASR (c) CIFAR-10 + Acc (d) CIFAR-10 + ASR

Figure 4: Accuracy and ASR as the unlearn-client ratio increases from 10% to 50%.

**Robustness to forgetting ratio.** Fig. 4 increases the fraction of clients to be forgotten, creating a larger deletion request. **FU-Dual** remains strong on both datasets, preserving useful accuracy while keeping ASR controlled. FCU, PGD, and MoDe are evaluated in their original single-target-client setting, since multi-client extensions are not directly defined and would introduce additional design choices. More details about varying the number of clients are provided in Appendix C.

**Robustness to distillation steps.** Fig. 5 varies the proxy-distillation budget. Accuracy improves quickly and then saturates, while ASR remains low around the main-report setting. Increasing the budget further brings only marginal change. Fig. 5 also shows that proxy quality has diminishing returns for unlearning. After the early improvement stage, further optimization brings little additional benefit, suggesting that FU-Dual only needs sufficiently informative proxy data rather than exhaustively optimized distilled data.

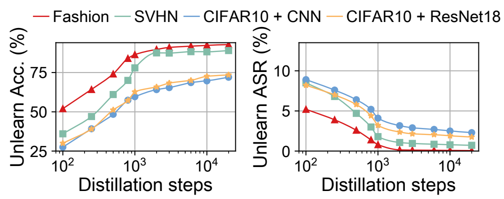


Figure 5: Effect of proxy-data distillation steps on unlearning performance.

### 5.4 Ablation Study

**Ablation on Dual Mapping.** Table 4 isolates the two logit mappings under the same proxy data and unlearning protocol. The full dual-mapping variant performs best across all reported datasets, while each single mapping gives only partial improvement.

Table 4: **Dual mapping ablation**.

| **Dataset** | **None** | $\mathcal{T}_{\mathcal{D}}$ | $\mathcal{T}_{\mathcal{U}}$ | **Dual** |
|---|---|---|---|---|
| SVHN | 82.43 | 83.54 | 85.84 | **87.52** |
| F-MNIST | 72.12 | 76.61 | 80.43 | **89.74** |
| CIFAR-10 | 57.41 | 61.52 | 61.61 | **64.13** |

Table 5: **Mapping method ablation**.

| Map | CIFAR-10 | | Fashion-MNIST | | SVHN | |
|---|---|---|---|---|---|---|
| | Distill | Unlearn | Distill | Unlearn | Distill | Unlearn |
| Linear | 85.20 | 87.10 | 53.20 | 81.10 | 60.40 | 92.40 |
| MLP | 87.78 | 89.90 | 57.10 | 84.82 | 64.18 | 94.68 |

**Ablation on mapping methods.** Table 5 compares linear and MLP mappings in the main text, while the complete Top-1, cosine-similarity, and KL-divergence results for LR, polynomial, and MLP mappings are reported in Appendix B. With the unlearning pipeline fixed, the MLP mapper gives stronger logit alignment than the linear mapper across both distillation and unlearning mappings. This suggests that the relation between proxy-side and client-side logits is not purely linear, and that a lightweight nonlinear mapper can better capture the functional correspondence needed by FU-Dual. The consistent advantage across additional metrics further indicates that the improvement is not tied to a single evaluation criterion, but reflects more faithful cross-model alignment.

**5.5 Discussion on Privacy** FU-Dual uses distilled proxies as functional supervision rather than as a replay buffer of private samples. Appendix D reports MIA leakage, FID-based separation, and qualitative results, all showing no observed privacy risk from the proxy data in our experiments.

## 6 Conclusion

We propose FU-Dual, a client-free framework that addresses reliance on client re-participation by shifting unlearning from parameter to logit space. By synergizing client-side data distillation with server-side knowledge distillation, our novel dual logit mappings provide precise guidance for the unlearning process. Specifically, these mappings serve as a dual-objective compass: one simulates the output shift required to erase client influence, while the other bridges the distributional gap between synthetic and real data. This approach effectively utilizes compact synthetic data, eliminating the need for any historical information or client re-participation. Extensive empirical evaluations demonstrate that FU-Dual achieves a superior balance between high utility retention and precise unlearning efficacy, significantly outperforming baselines without requiring any client re-participation. Furthermore, the robustness of FU-Dual is further verified across diverse and challenging scenarios.

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## Appendix Overview

The appendix provides supporting material for the theoretical, empirical, and reproducibility claims made in the main paper. Appendix A presents the theoretical analysis behind the dual-mapping design. It first studies the unlearning mapping and bounds the discrepancy between the ideal real-data mapping and its proxy-data estimate, then analyzes the distillation mapping, and finally explains why dataset distillation can support the proxy-data assumption used by FU-Dual.

The remaining appendices expand the empirical evidence and experimental protocol. Appendix B reports complete mapping-function ablations, while Appendix C evaluates robustness as the federation size changes. The post-unlearning fine-tuning section further examines whether each unlearned model provides a useful initialization for subsequent recovery. Appendix D studies the privacy characteristics of distilled proxy data through qualitative visualizations and quantitative leakage measurements. Appendix E provides the full comparison table, and the baseline-setting, computation-detail, and experimental-setup sections document the hyperparameters, cost accounting, datasets, metrics, and hardware needed to reproduce the experiments.

## A Theoretical Analysis and Proofs

### A.1 Unlearning Mapping Analysis

**Theorem A.1** (Explicit First-Order Error Bound of the Unlearning Mapping)**.** *Let $\hat{\mathcal{D}}_i \subset \mathcal{D}_i$ be an independently and uniformly sampled subset of client $i$, and suppose that all clients use the same sampling ratio*

$$\rho = \frac{|\hat{\mathcal{D}}_i|}{|\mathcal{D}_i|}, \qquad \forall i \in \mathcal{N} \cup \mathcal{R}. \tag{14}$$

*For each client $i$, let $\theta_i$ and $\hat{\theta}_i$ be the linear regression parameters trained on $\mathcal{D}_i$ and $\hat{\mathcal{D}}_i$, respectively. Let $\theta_{\mathcal{N}}, \theta_{\mathcal{R}}$ and $\hat{\theta}_{\mathcal{N}}, \hat{\theta}_{\mathcal{R}}$ denote the corresponding group-level parameters. The ideal and sampled unlearning mappings are defined by*

$$\phi^* = \theta_{\mathcal{N}}^{\dagger}\theta_{\mathcal{R}}, \qquad \phi = \hat{\theta}_{\mathcal{N}}^{\dagger}\hat{\theta}_{\mathcal{R}}. \tag{15}$$

*Assume that $\theta_{\mathcal{N}}$ and $\hat{\theta}_{\mathcal{N}}$ have the same rank and satisfy*

$$\sigma_{\min}^{+}(\theta_{\mathcal{N}}) \geq \gamma, \qquad \sigma_{\min}^{+}(\hat{\theta}_{\mathcal{N}}) \geq \gamma \tag{16}$$

*for some $\gamma > 0$. Assume further that*

$$\|\hat{\theta}_{\mathcal{N}} - \theta_{\mathcal{N}}\|_2^2 \leq \sum_{i\in\mathcal{N}} \|\hat{\theta}_i - \theta_i\|_2^2, \qquad \|\hat{\theta}_{\mathcal{R}} - \theta_{\mathcal{R}}\|_2^2 \leq \sum_{i\in\mathcal{R}} \|\hat{\theta}_i - \theta_i\|_2^2. \tag{17}$$

*For each client $i$, write*

$$\mathcal{D}_i = \{(x_i^k, y_i^k)\}_{k=1}^{m_i}, \qquad m_i = |\mathcal{D}_i|, \tag{18}$$

*and let $D_i$ be the design matrix of $\mathcal{D}_i$. Denote the full-data residual by*

$$\epsilon_i^k = y_i^k - f(x_i^k;\theta_i). \tag{19}$$

*Then the first-order mean squared error of the unlearning mapping satisfies*

$$\begin{aligned}
\mathbb{E}\|\phi - \phi^*\|_2^2 &\leq \xi \left[\sum_{i\in\mathcal{N}} \mathbb{E}\|\theta_i - \hat{\theta}_i\|_2^2 + \sum_{i\in\mathcal{R}} \mathbb{E}\|\theta_i - \hat{\theta}_i\|_2^2\right] \\
&\leq \xi \left[\sum_{i\in\mathcal{N}} \mathbb{E}\|\frac{f(X;\theta_i) - f(X;\hat{\theta}_i)}{\|X\|_2^2}\|_2^2 + \sum_{i\in\mathcal{R}} \mathbb{E}\|\frac{f(X;\theta_i) - f(X;\hat{\theta}_i)}{\|X\|_2^2}\|_2^2\right] \\
&\leq \frac{1-\rho}{\rho}\left[\frac{(1+\sqrt{5})^2\|\theta_{\mathcal{R}}\|_2^2}{2\gamma^4} \sum_{i\in\mathcal{N}} \frac{1}{m_i(m_i-1)} \sum_{k=1}^{m_i} \left\| \left(\frac{1}{m_i} D_i^{\top} D_i\right)^{-1} x_i^k \epsilon_i^k \right\|_2^2 \right. \\
&\qquad \left. + \frac{2}{\gamma^2} \sum_{i\in\mathcal{R}} \frac{1}{m_i(m_i-1)} \sum_{k=1}^{m_i} \left\| \left(\frac{1}{m_i} D_i^{\top} D_i\right)^{-1} x_i^k \epsilon_i^k \right\|_2^2 \right].
\end{aligned} \tag{20}$$

*Consequently,*

$$\mathbb{E}\|\phi - \phi^*\|_2^2 = O\left(\frac{1-\rho}{\rho}\right), \tag{21}$$

*where the hidden constant is independent of $\rho$.*

*Proof.* By the definition of the unlearning mappings,

$$\phi - \phi^* = \hat{\theta}_{\mathcal{N}}^{\dagger}\hat{\theta}_{\mathcal{R}} - \theta_{\mathcal{N}}^{\dagger}\theta_{\mathcal{R}}. \tag{22}$$

Decomposing the difference gives

$$\begin{aligned}\|\phi - \phi^*\|_2 &= \left\|\hat{\theta}_{\mathcal{N}}^{\dagger}(\hat{\theta}_{\mathcal{R}} - \theta_{\mathcal{R}}) + (\hat{\theta}_{\mathcal{N}}^{\dagger} - \theta_{\mathcal{N}}^{\dagger})\theta_{\mathcal{R}}\right\|_2 \\ &\leq \|\hat{\theta}_{\mathcal{N}}^{\dagger}\|_2\|\hat{\theta}_{\mathcal{R}} - \theta_{\mathcal{R}}\|_2 + \|\hat{\theta}_{\mathcal{N}}^{\dagger} - \theta_{\mathcal{N}}^{\dagger}\|_2\|\theta_{\mathcal{R}}\|_2.\end{aligned} \tag{23}$$

Using $(a+b)^2 \leq 2a^2 + 2b^2$, we obtain

$$\|\phi - \phi^*\|_2^2 \leq 2\|\hat{\theta}_{\mathcal{N}}^{\dagger}\|_2^2\|\hat{\theta}_{\mathcal{R}} - \theta_{\mathcal{R}}\|_2^2 + 2\|\hat{\theta}_{\mathcal{N}}^{\dagger} - \theta_{\mathcal{N}}^{\dagger}\|_2^2\|\theta_{\mathcal{R}}\|_2^2. \tag{24}$$

Since

$$\|\hat{\theta}_{\mathcal{N}}^{\dagger}\|_2 = \frac{1}{\sigma_{\min}^{+}(\hat{\theta}_{\mathcal{N}})} \leq \frac{1}{\gamma}, \tag{25}$$

and since $\theta_{\mathcal{N}}$ and $\hat{\theta}_{\mathcal{N}}$ have the same rank, the Moore–Penrose pseudoinverse perturbation bound yields

$$\begin{aligned}\|\hat{\theta}_{\mathcal{N}}^{\dagger} - \theta_{\mathcal{N}}^{\dagger}\|_2 &\leq \frac{1+\sqrt{5}}{2}\max\left\{\|\hat{\theta}_{\mathcal{N}}^{\dagger}\|_2^2, \|\theta_{\mathcal{N}}^{\dagger}\|_2^2\right\}\|\hat{\theta}_{\mathcal{N}} - \theta_{\mathcal{N}}\|_2 \\ &\leq \frac{1+\sqrt{5}}{2\gamma^2}\|\hat{\theta}_{\mathcal{N}} - \theta_{\mathcal{N}}\|_2.\end{aligned} \tag{26}$$

Therefore,

$$\|\phi - \phi^*\|_2^2 \leq \frac{2}{\gamma^2}\|\hat{\theta}_{\mathcal{R}} - \theta_{\mathcal{R}}\|_2^2 + \frac{(1+\sqrt{5})^2\|\theta_{\mathcal{R}}\|_2^2}{2\gamma^4}\|\hat{\theta}_{\mathcal{N}} - \theta_{\mathcal{N}}\|_2^2. \tag{27}$$

Taking expectation and applying the assumed group-level perturbation bounds gives

$$\mathbb{E}\|\phi - \phi^*\|_2^2 \leq \frac{(1+\sqrt{5})^2\|\theta_{\mathcal{R}}\|_2^2}{2\gamma^4}\sum_{i\in\mathcal{N}}\mathbb{E}\|\hat{\theta}_i - \theta_i\|_2^2 + \frac{2}{\gamma^2}\sum_{i\in\mathcal{R}}\mathbb{E}\|\hat{\theta}_i - \theta_i\|_2^2. \tag{28}$$

It remains to evaluate the first-order sampling error of the client-level linear regression estimators. For client $i$, the least-squares normal equation on the full data implies

$$\sum_{k=1}^{m_i} x_i^k \epsilon_i^k = 0. \tag{29}$$

The first-order perturbation induced by replacing $\mathcal{D}_i$ with a uniformly sampled subset $\hat{\mathcal{D}}_i$ is therefore governed by

$$\left(\frac{1}{m_i}D_i^{\top}D_i\right)^{-1} x_i^k\epsilon_i^k. \tag{30}$$

Since $|\hat{\mathcal{D}}_i| = \rho m_i$, the finite-population variance formula for sampling without replacement gives

$$\mathbb{E}\|\hat{\theta}_i - \theta_i\|_2^2 \leq \frac{1-\rho}{\rho}\frac{1}{m_i(m_i-1)}\sum_{k=1}^{m_i}\left\|\left(\frac{1}{m_i}D_i^{\top}D_i\right)^{-1}x_i^k\epsilon_i^k\right\|_2^2. \tag{31}$$

Substituting this bound into the previous inequality yields

$$\begin{aligned}\mathbb{E}\|\phi - \phi^*\|_2^2 \leq \frac{1-\rho}{\rho}\Bigg[&\frac{(1+\sqrt{5})^2\|\theta_{\mathcal{R}}\|_2^2}{2\gamma^4}\sum_{i\in\mathcal{N}}\frac{1}{m_i(m_i-1)}\sum_{k=1}^{m_i}\left\|\left(\frac{1}{m_i}D_i^{\top}D_i\right)^{-1}x_i^k\epsilon_i^k\right\|_2^2 \\ &+ \frac{2}{\gamma^2}\sum_{i\in\mathcal{R}}\frac{1}{m_i(m_i-1)}\sum_{k=1}^{m_i}\left\|\left(\frac{1}{m_i}D_i^{\top}D_i\right)^{-1}x_i^k\epsilon_i^k\right\|_2^2\Bigg].\end{aligned} \tag{32}$$

All terms inside the brackets are determined by the full-data design matrices, the residual vectors, and the singular-value lower bound $\gamma$, and are independent of the sampling ratio $\rho$. Hence,

$$\mathbb{E}\|\phi - \phi^*\|_2^2 = O\left(\frac{1-\rho}{\rho}\right). \tag{33}$$

This completes the proof. □

**A.2 Distillation Mapping Analysis** We analyze the transferability of the distillation mapping. The server learns the computable all-client distillation mapping $\mathcal{T}_{D,\mathcal{N}} : f(\mathcal{D};\hat{\theta}_{\mathcal{N}}) \to f(\mathcal{D};\theta_{\mathcal{N}})$, while the ideal utility-preserving mapping after unlearning should be $\mathcal{T}^*_{D,\mathcal{R}} : f(\mathcal{D};\hat{\theta}_{\mathcal{R}}) \to f(\mathcal{D};\theta_{\mathcal{R}})$. Thus, the goal is to bound the gap between these two mappings.

**Lemma A.2.** *Let $f(X;\theta) = X\theta$ be a linear model. The all-client and remaining-client distillation mappings can be parameterized by $\psi_{\mathcal{N}} = \hat{\theta}^{\dagger}_{\mathcal{N}}\theta_{\mathcal{N}}$ and $\psi^*_{\mathcal{R}} = \hat{\theta}^{\dagger}_{\mathcal{R}}\theta_{\mathcal{R}}$, respectively.*

*Proof.* For linear models, a least-squares logit-space mapping from $f(X;\hat{\theta}_{\mathcal{N}})$ to $f(X;\theta_{\mathcal{N}})$ is given by $\psi_{\mathcal{N}} = \hat{\theta}^{\dagger}_{\mathcal{N}}\theta_{\mathcal{N}}$. Similarly, the ideal remaining-client proxy-real mapping from $f(X;\hat{\theta}_{\mathcal{R}})$ to $f(X;\theta_{\mathcal{R}})$ is given by $\psi^*_{\mathcal{R}} = \hat{\theta}^{\dagger}_{\mathcal{R}}\theta_{\mathcal{R}}$. Therefore, $\psi_{\mathcal{N}}$ is the computable all-client distillation mapping and $\psi^*_{\mathcal{R}}$ is the ideal remaining-client distillation mapping. □

**Theorem A.3** (Bounded Transfer Error of the Distillation Mapping)**.** *Let $f(X;\theta) = X\theta$ be a linear model. Define the computable all-client distillation mapping and the ideal remaining-client distillation mapping as $\psi_{\mathcal{N}} = \hat{\theta}^{\dagger}_{\mathcal{N}}\theta_{\mathcal{N}}$ and $\psi^*_{\mathcal{R}} = \hat{\theta}^{\dagger}_{\mathcal{R}}\theta_{\mathcal{R}}$, respectively. Under the same linear-case perturbation regularity used in the unlearning mapping analysis, assume $\hat{\theta}^{\dagger}_{\mathcal{N}}\hat{\theta}_{\mathcal{N}} = I$ and $\hat{\theta}^{\dagger}_{\mathcal{R}}\hat{\theta}_{\mathcal{R}} = I$. Then, under Assumption 3.3, there exists a positive constant $\xi_D$ such that*

$$\mathbb{E}\|\psi_{\mathcal{N}} - \psi^*_{\mathcal{R}}\|_F^2 \le \xi_D\left[\sum_{i\in\mathcal{N}} C_i^2 + \sum_{i\in\mathcal{R}} C_i^2\right]. \tag{34}$$

*Here, $\xi_D$ denotes the resulting positive conditioning constant that collects the pseudoinverse norm factors and the fixed constants in the group-level and output-to-parameter reductions; it is not an additional proxy-quality assumption.*

*Proof.* By the definitions of the computable and ideal distillation mappings,

$$\mathbb{E}\|\psi_{\mathcal{N}} - \psi^*_{\mathcal{R}}\|_F^2 = \mathbb{E}\left\|\hat{\theta}^{\dagger}_{\mathcal{N}}\theta_{\mathcal{N}} - \hat{\theta}^{\dagger}_{\mathcal{R}}\theta_{\mathcal{R}}\right\|_F^2. \tag{35}$$

Using $\hat{\theta}^{\dagger}_{\mathcal{N}}\hat{\theta}_{\mathcal{N}} = I$ and $\hat{\theta}^{\dagger}_{\mathcal{R}}\hat{\theta}_{\mathcal{R}} = I$, we have

$$\psi_{\mathcal{N}} - \psi^*_{\mathcal{R}} = \hat{\theta}^{\dagger}_{\mathcal{N}}(\theta_{\mathcal{N}} - \hat{\theta}_{\mathcal{N}}) - \hat{\theta}^{\dagger}_{\mathcal{R}}(\theta_{\mathcal{R}} - \hat{\theta}_{\mathcal{R}}). \tag{36}$$

Using $\|a-b\|_F^2 \le 2\|a\|_F^2 + 2\|b\|_F^2$, and absorbing the conditioning constants of the pseudoinverses into $\xi_D$, we obtain

$$\mathbb{E}\|\psi_{\mathcal{N}} - \psi^*_{\mathcal{R}}\|_F^2 \le \xi_D\left[\mathbb{E}\|\theta_{\mathcal{N}} - \hat{\theta}_{\mathcal{N}}\|_F^2 + \mathbb{E}\|\theta_{\mathcal{R}} - \hat{\theta}_{\mathcal{R}}\|_F^2\right]. \tag{37}$$

Here, $\xi_D$ is chosen large enough to dominate the pseudoinverse factors $2\|\hat{\theta}^{\dagger}_{\mathcal{N}}\|_2^2$ and $2\|\hat{\theta}^{\dagger}_{\mathcal{R}}\|_2^2$, together with the fixed conditioning constants used when reducing group-level parameter discrepancies to client-level normalized output discrepancies. Following the same group-level decomposition used in the unlearning mapping analysis, the group-level proxy-real discrepancies are controlled by client-level discrepancies:

$$\mathbb{E}\|\theta_{\mathcal{N}} - \hat{\theta}_{\mathcal{N}}\|_F^2 + \mathbb{E}\|\theta_{\mathcal{R}} - \hat{\theta}_{\mathcal{R}}\|_F^2 \le \sum_{i\in\mathcal{N}} \mathbb{E}\|\theta_i - \hat{\theta}_i\|_F^2 + \sum_{i\in\mathcal{R}} \mathbb{E}\|\theta_i - \hat{\theta}_i\|_F^2. \tag{38}$$

For linear models, the client-level parameter discrepancy is controlled by the normalized output discrepancy, yielding

$$\mathbb{E}\|\theta_i - \hat{\theta}_i\|_F^2 \le \mathbb{E}\left\|\frac{f(X;\theta_i) - f(X;\hat{\theta}_i)}{\|X\|_2^2}\right\|_2^2. \tag{39}$$

Substituting the above inequality gives

$$\mathbb{E}\|\psi_{\mathcal{N}} - \psi^*_{\mathcal{R}}\|_F^2 \le \xi_D \left[ \sum_{i\in\mathcal{N}} \mathbb{E} \left\| \frac{f(X;\theta_i) - f(X;\hat{\theta}_i)}{\|X\|_2^2} \right\|_2^2 + \sum_{i\in\mathcal{R}} \mathbb{E} \left\| \frac{f(X;\theta_i) - f(X;\hat{\theta}_i)}{\|X\|_2^2} \right\|_2^2 \right]. \quad (40)$$

By Assumption 3.3, $\left\|(f(X;\theta_i) - f(X;\hat{\theta}_i))/\|X\|_2^2\right\|_2 \le C_i$, and hence

$$\mathbb{E} \left\| \frac{f(X;\theta_i) - f(X;\hat{\theta}_i)}{\|X\|_2^2} \right\|_2^2 \le C_i^2. \quad (41)$$

Therefore,

$$\mathbb{E}\|\psi_{\mathcal{N}} - \psi^*_{\mathcal{R}}\|_F^2 \le \xi_D \left[ \sum_{i\in\mathcal{N}} C_i^2 + \sum_{i\in\mathcal{R}} C_i^2 \right]. \quad (42)$$

This completes the proof. □

*Remark* A.4. The above result gives a bounded-error guarantee rather than a convergence rate. The bound depends on the proxy-real approximation quality specified in Assumption 3.3. Therefore, when the distilled proxy data induce models whose outputs are close to those of real-data models, the all-client distillation mapping $\psi_{\mathcal{N}}$ remains close to the ideal remaining-client mapping $\psi^*_{\mathcal{R}}$. This supports applying the mapping learned from $f(\mathcal{D};\hat{\theta}_{\mathcal{N}}) \to f(\mathcal{D};\theta_{\mathcal{N}})$ to $f(\mathcal{D};\hat{\theta}_{\mathcal{R}})$ during server-side unlearning.

**A.3 Dataset Distillation and the Proxy Assumption** This section expands the justification for Assumption 3.3. In the standard dataset distillation formulation, the $i$-th client optimizes a compact distilled dataset $\hat{\mathcal{D}}_i$ and a learning rate $\hat{\eta}_i$ such that one update on $\hat{\mathcal{D}}_i$ yields low loss on the original local dataset $\mathcal{D}_i$:

$$\hat{\mathcal{D}}_i^*, \hat{\eta}_i^* = \arg\min_{\hat{\mathcal{D}}_i, \hat{\eta}_i} \ell\left(\mathcal{D}_i, \theta_{i,0} - \hat{\eta}_i \nabla_{\theta_{i,0}} \ell(\hat{\mathcal{D}}_i, \theta_{i,0})\right). \quad (43)$$

This objective requires the distilled data to preserve the training effect of the original local data. Equivalently, the prediction-level view of dataset distillation [Yu et al., 2024] optimizes $\hat{\mathcal{D}}_i$ so that the model trained on distilled data matches the model trained on real data over the local distribution:

$$\min_{\hat{\mathcal{D}}_i} \int p_i(x)\, d\Big(f(x;\theta_i), f(x;\hat{\theta}_i)\Big)\, dx. \quad (44)$$

Here, $p_i(x)$ is the class-conditional local distribution, i.e., $p_i(x) \triangleq p_i(x \mid y)$ for samples with label $y$, $d(\cdot,\cdot)$ is a prediction discrepancy, and $\theta_i$ and $\hat{\theta}_i$ are obtained by training the same architecture on $\mathcal{D}_i$ and $\hat{\mathcal{D}}_i$, respectively.

This view shows that Assumption 3.3 is not introduced as an independent model-closeness requirement. Rather, when the prediction-matching objective is optimized well, the pointwise residual can be controlled by a small optimization-dependent quantity $\gamma_i$, i.e., $\|f(x;\theta_i) - f(x;\hat{\theta}_i)\| \le \gamma_i$. For inputs with non-vanishing magnitude $\|x\|_2 \ge r_i > 0$, this gives

$$\frac{\left\|f(X;\theta_i) - f(X;\hat{\theta}_i)\right\|}{\|X\|_2^2} \le \frac{\gamma_i}{\|X\|_2^2} \le \frac{\gamma_i}{r_i^2} := C_i. \quad (45)$$

Therefore, a well-optimized distilled dataset naturally yields the input-scale-adjusted output discrepancy used in Assumption 3.3. The tolerance $C_i$ is determined by the achieved prediction-matching residual and the input scale, rather than by an additional absolute closeness assumption between the two trained models.

# B Ablation Details

Table 6 evaluates whether the dual mappings need nonlinear capacity. Across all datasets and both mapping types, the MLP mapper gives the best Top-1 agreement, cosine similarity, and KL divergence

Table 6: **Mapping metrics by dataset and mapping function.** D and U denote the distillation and unlearning mappings, respectively; higher Top-1/Cos. and lower KL indicate better logit alignment.

| Mapping | Metric | CIFAR-10 | | Fashion-MNIST | | SVHN | |
|---|---|---|---|---|---|---|---|
| | | D | U | D | U | D | U |
| LR | Top-1 ↑ | 86.90 | 79.80 | 44.80 | 67.50 | 59.50 | 94.20 |
| | Cos. ↑ | 85.20 | 87.10 | 53.20 | 81.10 | 60.40 | 92.40 |
| | $\mathrm{KL}_{p\|q}$ ↓ | 0.420 | 0.610 | 5.100 | 1.350 | 5.300 | 0.120 |
| Poly | Top-1 ↑ | 88.20 | 81.10 | 46.90 | 69.80 | 61.80 | 95.10 |
| | Cos. ↑ | 86.90 | 88.90 | 55.60 | 83.40 | 62.80 | 93.90 |
| | $\mathrm{KL}_{p\|q}$ ↓ | 0.360 | 0.550 | 4.700 | 1.220 | 4.900 | 0.105 |
| MLP | Top-1 ↑ | 89.44 | 82.38 | 48.38 | 71.27 | 63.05 | 95.94 |
| | Cos. ↑ | 87.78 | 89.90 | 57.10 | 84.82 | 64.18 | 94.68 |
| | $\mathrm{KL}_{p\|q}$ ↓ | 0.330 | 0.516 | 4.453 | 1.161 | 4.690 | 0.099 |

among the tested alternatives. The gain is consistent for the unlearning mapping, where the mapper must represent the shift from all-client behavior to remaining-client behavior, and for the distillation mapping, where it must calibrate proxy-domain logits to real-model logits. This supports using a lightweight nonlinear mapper rather than restricting the transformation to a linear or polynomial form.

## C Client-Scaling Robustness

Fig. 6 reports the robustness experiment under different federation sizes. We vary the total number of clients from 10 to 60 and report the accuracy drop relative to the corresponding FL model before unlearning: $\Delta\mathrm{Acc} = \mathrm{Acc}_{\mathrm{FL}} - \mathrm{Acc}_{\mathrm{unlearn}}$. Thus, smaller bars indicate that an unlearning method better preserves the utility of the original FL model at the same federation size.

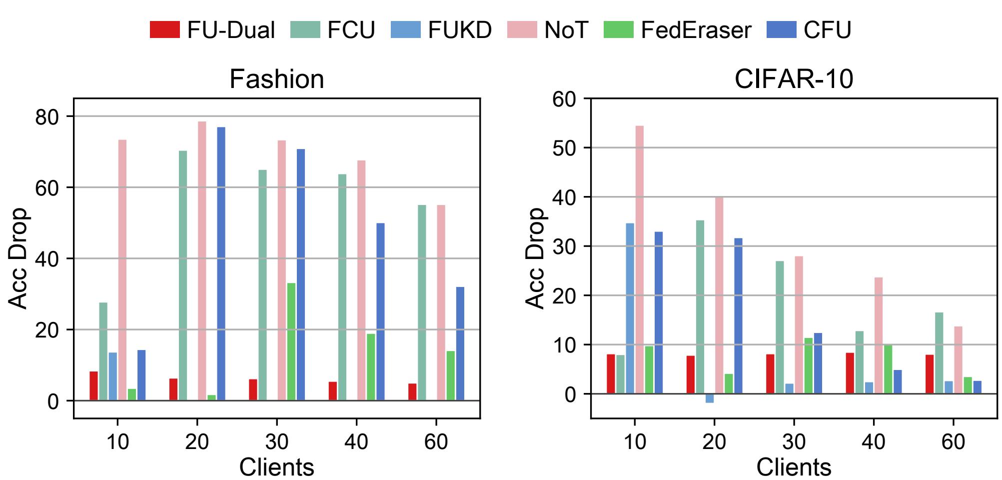
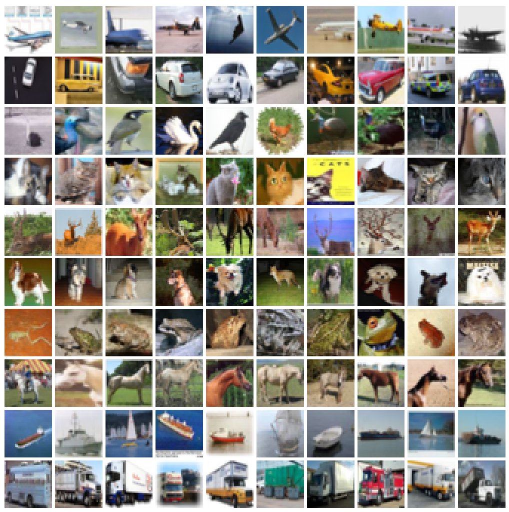



Figure 6: Accuracy drop relative to the corresponding FL model under different federation sizes. Lower is better.

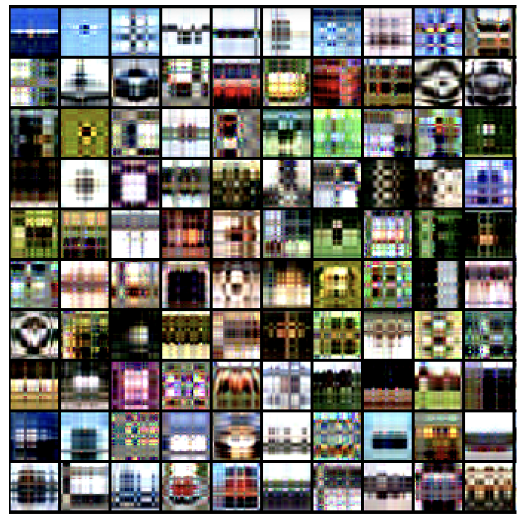
The main pattern is that FU-Dual shows stable utility retention as the federation grows. On Fashion-MNIST, its drop stays within a narrow range from 4.9 to 8.3 percentage points across all tested client counts. By contrast, several peer methods show pronounced utility loss or instability: FCU, NoT, and CFU often lose more than 50 points, and FedEraser becomes less stable once the federation size increases beyond 20 clients. On CIFAR-10, FU-Dual remains similarly stable, with drops between

7.8 and 8.4 points from 10 to 60 clients. In contrast, FCU and NoT incur much larger utility loss, while FedEraser and CFU improve at larger client counts but are less consistent across the full range. This utility-retention view complements the unlearn-ratio study in Fig. 4: FU-Dual remains robust not only when more clients are forgotten, but also when the overall federation size changes.

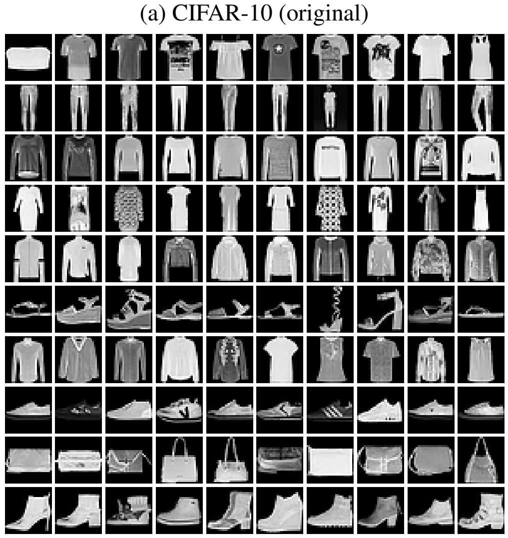
# D Privacy Analysis



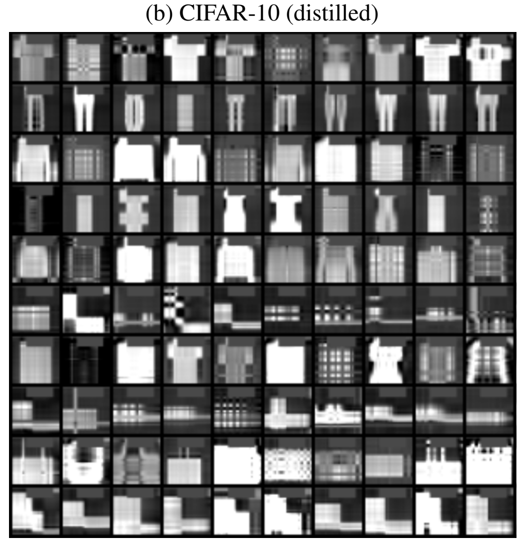
(a) CIFAR-10 (original)



(b) CIFAR-10 (distilled)



(c) Fashion-MNIST (original)



(d) Fashion-MNIST (distilled)

Figure 7: **Privacy comparison before and after distillation.** For each dataset, the left image shows original training samples and the right image shows the corresponding distilled proxy samples.

**D.1 Qualitative Privacy Analysis: Visual Privacy Characteristics** In Fig. 7a and 7b, as well as Fig. 7c and 7d, we visualize the LoDM-distilled images for CIFAR-10 and Fashion-MNIST. Compared with the original images, the distilled samples are abstract and do not preserve clear instance-level visual details. This qualitative difference is not used as a stand-alone privacy guarantee, but it is consistent with the quantitative measurements in Table 7: the proxies remain distributionally separated from the original data while supporting model behavior needed for unlearning.

**D.2 Quantitative Privacy Evaluation: Privacy Leakage Measurement** As shown in Table 7, FU-Dual demonstrates a robust privacy profile under the low Images Per Class (IPC) setting used

Table 7: **Privacy leakage and distributional separation of distilled proxies.** Lower $\mathrm{AUC}_{\mathrm{leak}}$ and TPR@1%FPR indicate weaker membership leakage, while larger FID indicates stronger visual/distributional separation from original training samples.

| Dataset | $\mathrm{AUC}_{\mathrm{leak}} \downarrow$ | TPR@1%FPR↓ | $\mathrm{AUC}_{\mathrm{ceil}}$ (ref.) | FID (mean ± std) |
|---|---|---|---|---|
| Fashion-MNIST | 0.504 | 0.010 | 0.561 | $93.79 \pm 0.84$ |
| CIFAR-10 | 0.510 | 0.011 | 0.690 | $217.67 \pm 1.77$ |
| SVHN | 0.498 | 0.013 | 0.563 | $236.66 \pm 5.13$ |

in our experiments. The leakage AUC remains close to 0.5 and TPR@1%FPR remains low across SVHN, Fashion-MNIST, and CIFAR-10, indicating limited membership leakage. The FID values further show that the distilled datasets are distributionally separated from their corresponding original training sets, supporting the qualitative evidence in Fig. 7. While increasing IPC may naturally increase privacy leakage risk, high-IPC configurations are counter-intuitive for data distillation. As the volume of synthetic images approaches that of the original dataset, the fundamental advantage of dataset condensation is effectively negated. Consequently, deploying data distillation in high-IPC settings is both impractical and atypical.

# E Full Main Results

Table 8: **Full main results across datasets and architectures.** Best results are **bold**, and second best are underlined (excluding Retrain).

| Dataset + Model | Method | Test Acc ↑ | ASR ↓ | MIA | Server Cost | | Client Cost | |
|---|---|---|---|---|---|---|---|---|
| | | | | | Comm. | Comp. | Comm. | Comp. |
| FashionMNIST + CNN | Retrain | 92.16±0.05% | 0±0.00% | 50.03±0.03% | 1.72e8 | 0 | 1.72e8 | 2.35e15 |
| | FedEraser | 87.36±0.37% | 0.43±0.31% | 50.52±0.42% | 6.86e7 | 0 | 6.86e7 | 1.31e15 |
| | FUKD | 77.15±0.45% | 3.21±0.40% | 53.66±0.48% | 1.72e8 | 1.74e13 | 1.72e8 | 3.54e15 |
| | FCU | 63.09±0.12% | 3.14±0.15% | 50.31±0.10% | 1.86e8 | 0 | 1.84e8 | 4.35e14 |
| | PGD | 87.58±0.18% | 98.47±0.22% | 57.43±0.16% | 2.54e6 | 0 | 1.27e6 | 1.30e13 |
| | MoDe | **89.78±0.20%** | 41.20±0.21% | 53.12±0.15% | 3.60e8 | 0 | 3.49e8 | 5.00e15 |
| | NoT | 17.32±0.09% | 3.16±0.11% | 49.26±0.08% | 1.14e7 | 0 | 1.14e7 | 0 |
| | FFMU | 80.95±0.10% | 0.60±0.12% | 50.17±0.09% | 0 | 0 | 0 | 0 |
| | CFU | 76.44±0.11% | 0.60±0.10% | 50.77±0.08% | 0 | 2.03e12 | 0 | 0 |
| | **FU-Dual (Ours)** | 89.74±0.07% | **0.15±0.04%** | **50.02±0.03%** | 0 | 4.70e13 | 0 | 0 |
| SVHN + CNN | Retrain | 89.97±0.06% | 0±0.00% | 50.02±0.03% | 1.73e8 | 0 | 1.73e8 | 3.01e15 |
| | FedEraser | 85.73±0.41% | **1.06±0.38%** | 50.31±0.34% | 6.91e7 | 0 | 6.91e7 | 1.67e15 |
| | FUKD | 71.91±0.48% | 50.63±0.45% | 53.36±0.43% | 1.73e8 | 4.76e13 | 1.73e8 | 4.55e15 |
| | FCU | 87.23±0.13% | 56.22±0.16% | 53.15±0.11% | 1.87e8 | 0 | 1.86e8 | 5.57e14 |
| | PGD | 84.23±0.19% | 51.98±0.23% | 53.20±0.14% | 2.56e6 | 0 | 1.28e6 | 1.60e13 |
| | MoDe | 83.41±0.18% | 74.35±0.24% | 53.24±0.17% | 3.62e8 | 0 | 3.52e8 | 6.41e15 |
| | NoT | 65.65±0.10% | 23.72±0.13% | 53.08±0.10% | 1.15e7 | 0 | 1.15e7 | 0 |
| | FFMU | 80.80±0.12% | 22.85±0.14% | 53.05±0.11% | 0 | 0 | 0 | 0 |
| | CFU | 69.41±0.11% | 2.99±0.10% | 50.98±0.09% | 0 | 2.13e12 | 0 | 0 |
| | **FU-Dual (Ours)** | **87.52±0.08%** | 1.07±0.05% | **50.13±0.04%** | 0 | 6.10e13 | 0 | 0 |
| CIFAR10 + CNN | Retrain | 72.21±0.07% | 0±0.00% | 50.00±0.02% | 1.73e8 | 0 | 1.73e8 | 2.06e15 |
| | FedEraser | 58.52±0.46% | 7.98±0.43% | 50.09±0.39% | 6.91e7 | 0 | 6.91e7 | 1.14e15 |
| | FUKD | 33.55±0.42% | **3.17±0.47%** | **50.06±0.36%** | 1.73e8 | 1.83e13 | 1.73e8 | 3.09e15 |
| | FCU | 60.32±0.14% | 78.81±0.17% | 50.81±0.12% | 1.87e8 | 0 | 1.86e8 | 3.80e14 |
| | PGD | 50.23±0.20% | 88.54±0.24% | 51.09±0.16% | 2.56e6 | 0 | 1.28e6 | 1.10e13 |
| | MoDe | 46.68±0.19% | 87.00±0.22% | 50.09±0.13% | 3.62e8 | 0 | 3.52e8 | 4.37e15 |
| | NoT | 13.76±0.08% | 7.39±0.12% | 49.71±0.09% | 1.15e7 | 0 | 1.15e7 | 0 |
| | FFMU | 31.68±0.13% | 22.82±0.15% | 51.71±0.10% | 0 | 0 | 0 | 0 |
| | CFU | 35.28±0.12% | 6.86±0.11% | 50.92±0.08% | 0 | 2.13e12 | 0 | 0 |
| | **FU-Dual (Ours)** | **64.13±0.09%** | **3.17±0.05%** | 50.08±0.04% | 0 | 6.00e13 | 0 | 0 |
| CIFAR10 + ResNet18 | Retrain | 76.33±0.08% | 0±0.00% | 50.01±0.03% | 2.68e10 | 0 | 2.68e10 | 1.14e16 |
| | FedEraser | 63.97±0.44% | 4.23±0.36% | 50.71±0.47% | 1.34e10 | 0 | 1.34e10 | 5.70e15 |
| | FUKD | 45.37±0.46% | **1.90±0.44%** | 50.10±0.41% | 0 | 6.66e13 | 6.70e8 | 0 |
| | FCU | 64.21±0.15% | 68.01±0.18% | 55.12±0.13% | 4.47e7 | 0 | 4.47e7 | 1.11e15 |
| | PGD | 53.88±0.21% | 71.52±0.25% | 53.14±0.17% | 8.94e7 | 0 | 4.47e7 | 5.55e13 |
| | MoDe | 43.91±0.20% | 69.22±0.23% | 50.63±0.14% | 2.23e9 | 0 | 2.23e9 | 5.88e14 |
| | NoT | 14.31±0.09% | 5.35±0.12% | 48.91±0.10% | 4.47e7 | 0 | 0 | 0 |
| | FFMU | 33.57±0.13% | 3.10±0.11% | 50.41±0.09% | 0 | 0 | 0 | 0 |
| | CFU | 33.50±0.12% | 5.53±0.12% | 51.28±0.10% | 0 | 2.33e13 | 0 | 0 |
| | **FU-Dual (Ours)** | **65.82±0.09%** | 2.37±0.06% | **50.02±0.04%** | 0 | 5.44e13 | 0 | 0 |

# F Baseline setting details

Table 9 summarizes the hyperparameters used in the communication and computation accounting. The accounting uses a unified cost-analysis setting with $K = 10$ clients, one forgotten target client, nine retained clients, and non-IID degree 0.8. For shared FL and retained-client retraining, we use 15 global epochs and normalize local training to 10 local epochs. Forward FLOPs are measured per sample, and one training update is counted as $3F_{\text{fwd}}$ to cover forward and backward propagation. For selected high-non-IID phase-2 counters in PGD, FCU, FedEraser, FUKD, and MoDe, the epoch or round counter is multiplied by 1.5 in the cost sheet. Post-unlearning fine-tuning and recovery phases are excluded from the fair unlearning-cost split unless explicitly stated. For CFU, we count the server-side anchor-subspace reconstruction cost and report the auxiliary KD recovery separately. For FFMU, the preparation is performed during training; because it does not run a separate post-request deletion-stage optimization, its non-FL unlearning FLOPs are zero.

Table 9: **Core hyperparameters used for communication and computation accounting.**

| **Method** | **Parameter** | **Value** |
|---|---|---|
| **Shared** | model | CompatibleConvNet_32x32_v3 |
| | num_clients / retained_clients | $10/9$ |
| | global_epochs | 15 |
| | local_epochs | 10 |
| | client_batch_size / test_batch_size | $256/256$ |
| | train multiplier | $3F_{\text{fwd}}$ |
| | $F_{\text{fwd}}$ (MNIST/Fashion) | 96.77 MFLOPs |
| | $F_{\text{fwd}}$ (CIFAR-10/SVHN) | 101.49 MFLOPs |
| | high non-IID cost factor | 1.5 for selected phase-2 counters |
| FCU | iterations | $100 \times 1.5$ |
| | learning_rate | $1.0 \times 10^{-5}$ |
| | batch_size | 64 |
| | mcu_temperature | 0.5 |
| | alpha | 1.0 |
| | beta | 0.1 |
| | gamma | 0 |
| | fgmp_threshold | 0.5 |
| | downgrade_factor | 0.5 |
| | teacher count | 2 |
| FUKD | method | approximate |
| | model_exclude global_epochs | $15 \times 1.5$ |
| | model_exclude local_epochs | 10 |
| | retained_clients | 9 |
| | KD epochs | $50 \times 1.5$ for cost normalization |
| | distill_ratio | 0.1 |
| | temperature | 4.0 |
| | alpha / beta | $0.7/0.3$ |
| | learning_rate | 0.01 |
| | teacher count | 1 |
| MoDe | unlearning_type | client |
| | lambda | 0.95 |
| | degradation_rounds | $5 \times 1.5$ |
| | max_rounds | $8 \times 1.5$ |
| | local_epochs | $10 \times 1.5$ |
| | update_frequency | 1 |
| | guidance_epochs | 3 |
| | guidance_lr | $5.0 \times 10^{-4}$ |
| | use_pseudo_labels | true |
| NoT | layer_selection | first_layer |
| | memory_recovery rounds | excluded from fair cost split |
| | memory_recovery local_epochs | excluded from fair cost split |
| | learning_rate | $1.0 \times 10^{-3}$ |
| | batch_size | 64 |
| | recovery_ratio | 0.8 |
| | oblivion_technique | first-layer weight negation |
| PGD | learning_rate | 0.01 |
| | l2_radius | 1.0 |
| | epochs | $5 \times 1.5$ |
| | batch_size | 32 |

Continued on next page

| Method | Parameter | Value |
|---|---|---|
| | gradient_clip_radius | 5.0 |
| CFU | global_epochs / local_epochs | $15/10$ |
| | aux_size | 256 |
| | aux_batch_size | 32 |
| | anchor_dim_k | 100 |
| | power_method / power_niter | false / 2 |
| | aux_dataset | CIFAR-100 |
| | KD recovery epochs | excluded / reported separately |
| | KD temperature / learning_rate | $4.0/0.001$ |
| FFMU | global_epochs / local_epochs | $15/10$ |
| | sigma | 0.1 |
| | lambda | $\sigma^2$ |
| | pcmu_learning_rate | 0.1 |
| | aggregation | FedAvg |
| | bn_lr | 0 |
| | unlearning_stage_epochs | 0 |
| **FU-Dual** | IPC | 20 |
| | data-matching iterations | 1500 |
| | batch_real | 32 |
| | $E_{\text{mamb}}$ (MNIST/Fashion) | 20 |
| | $E_{\text{mamb}}$ (CIFAR-10/SVHN) | 30 |
| | $E_{\text{KD}}$ | 50 |
| | base set size $n_{\text{base}}$ | $20 \times 10 \times 9 = 1800$ |
| Retrain | retained_clients | 9 |
| | global_epochs | 15 |
| | local_epochs | 10 |
| | lr_local | 0.01 |

# G Computation details

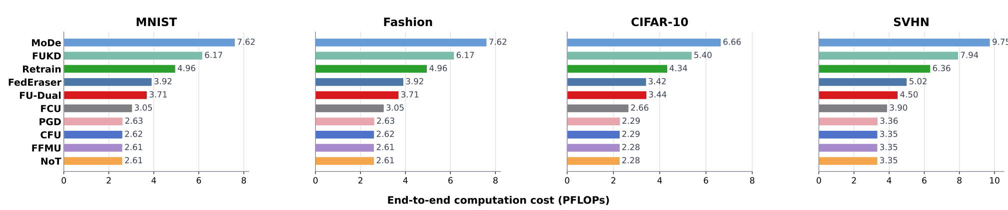



Figure 8: **End-to-end computation cost.** The all-stage view includes shared FL cost and, for FU-Dual, the one-time proxy data-matching stage in addition to mapping and student distillation.

Table 9 summarizes the hyperparameters used for the computation accounting, so we do not repeat the method-specific settings here. Fig. 8 provides an overview of the end-to-end computation cost, including ordinary FL training and the additional costs introduced by FU-Dual. For FU-Dual, this end-to-end view also includes proxy-data preparation, proxy reference training, logit mapping, and student distillation. Adding proxy data does not make the overall computation excessive, because the proxy bases are compact, the mappings are lightweight, and the post-request distillation is performed only on the proxy data rather than on original client datasets. The proxy data-matching stage is also a reusable preparation cost, which further reduces the amortized cost for later unlearning requests. Thus, FU-Dual remains computationally practical while avoiding client re-participation during unlearning.

# H Detailed Experimental Setup and Evaluation Metrics

**H.1 Experimental Setup** We evaluate **FU-Dual** and all baseline methods in two stagesThe experiments are conducted on Fashion-MNIST, SVHN, and CIFAR-10 with two model architectures, CNN and ResNet-18. For the main comparison, each dataset is partitioned across $K = 10$ clients under a highly non-IID federated setting, with the non-IID degree set to 0.8. Each client trains only on its private local split $\mathcal{D}_i$, while the global test set is used to evaluate utility after unlearning. Additional robustness studies vary the non-IID degree, the unlearn ratio, and the number of clients.

All experiments are conducted within a federated learning framework combined with dataset distillation. For the communication and computation accounting, the shared FL schedule uses 15 global epochs and 10 local epochs under the unified cost-analysis setting. Local models are trained using SGD with a learning rate of 0.01, weight decay 0, a client batch size of 256, a test batch size of 256.

For dataset distillation, each client constructs a compact proxy dataset using LoDM with an image-per-class (IPC) budget of 20. During both unlearning and fine-tuning, we apply logit-level knowledge distillation using an $\ell_1$ loss.

**H.2 Evaluation Metrics** We evaluate unlearning methods across three dimensions: utility retention, forgetting effectiveness, and resource consumption.

**Utility retention.** Utility retention is measured by test accuracy, which reflects whether the unlearned model can preserve its predictive performance on the remaining data after removing the target knowledge.

**Forgetting effectiveness.** Forgetting effectiveness is assessed using Attack Success Rate (ASR) Zhang et al. [2024] and Membership Inference Attack (MIA). ASR measures the model's response to target triggers. A value close to zero indicates that the target knowledge has been effectively removed. MIA further evaluates whether the model still retains membership information about the forgotten data. A successful unlearning method should reduce the MIA accuracy to approximately 50%, making the attack no better than random guessing.

**Resource consumption.** Resource consumption is quantified by communication and computation costs under the accounting views described above. Communication is reported in Bytes and computation is reported in FLOPs or PFLOPs. Unless otherwise stated, the main unlearning split excludes shared FL and post-unlearning fine-tuning/recovery, while the end-to-end computation figure uses all-stage PFLOPs.

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